

Control of TT Motor Drive using C2000 Microcontroller

*Report submitted to the SASTRA Deemed to be University
as the requirement for the course*

EEE320R01 : DIGITAL CONTROLLER FOR POWER ELECTRONIC APPLICATIONS

Submitted by

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Bonafide Certificate

This is to certify that the report titled “**Control of TT Motor Drive using C2000 Microcontroller**” submitted as a requirement for the course, **EEE320R01 :PROJECT BASED ASSESSMENT** for B. Tech. Electrical and Electronics Engineering with specialization in Smart Grid and Electric Vehicles programme, is a bonafide record of the work done by **Mr.Siva guru C S(Reg. No. 127159046)**, **Mr.Arul jeffry A(Reg. No. 127159008)**, during the academic year 2025 -2026, in the **School of Electrical and Electronics Engineering**, under my supervision.

Signature of Project Supervisor:

Name with Affiliation :**Dr.Santhosh.T.K.,(SAP /EEE/ SEEE)**

Date : / 11/ 2025

Project *Vivavoce* held on

Examiner-I

Examiner-II



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Declaration

We declare that the report titled “**Control of TT Motor Drive using C2000 Microcontroller**” submitted by us is an original work done by us under the guidance of **Dr. Santhosh. T.K., SAP, School of Electrical and Electronics Engineering, SASTRA Deemed to be University** during the academic year 2025-26, in the **School of Electrical and Electronics Engineering**. The work is original and wherever We have used materials from other sources, We have given due credit and cited them in the text of the report. This report has not formed the basis for the award of any degree, diploma, associate-ship, fellowship or other similar title to any candidate of any University.

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Date :

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ABSTRACT

This project focuses on the design and implementation of an open-loop synchronous buck converter to step down a 12 V DC input to 5 V DC for driving a DC motor. The converter uses two MOSFETs driven by an IR gate driver, with PWM signals generated from a TI C2000 microcontroller. The system operates in open loop, where the output voltage and motor speed are directly controlled by adjusting the PWM duty cycle..

Specific Contribution

- designing and implementation of an open-loop synchronous buck converter.
- generation of PWM signals using a C2000 microcontroller to control a synchronous buck converter.

Specific Learning

- Learnt how to design a PCB
- learnt how to simulate using MATLAB
- Understood how to incorporate c2000 with power converters

Signature of the Guide

Student Reg. No: 127159008

Name of the Guide: Dr.Santhosh.T.K(SAP/EEE/SEEE)

Name: Arul jeffry A

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CHAPTER 1

Introduction to Buck-converter w

1.1 Power converters

A power converter is an electrical device for converting electrical energy between alternating current (AC) and direct current (DC). the voltage or frequency of the current can change. Based on the performance of the power conversion they perform power converters are classified. One way of classifying power conversion systems is based on whether the input and output is alternating or direct current,

1.2 Synchronous Buck converter

A synchronous buck converter is a type of DC-DC step-down converter that efficiently converts a higher input voltage into a lower regulated output voltage. It operates on the principle of high-frequency switching and energy storage in an inductor.

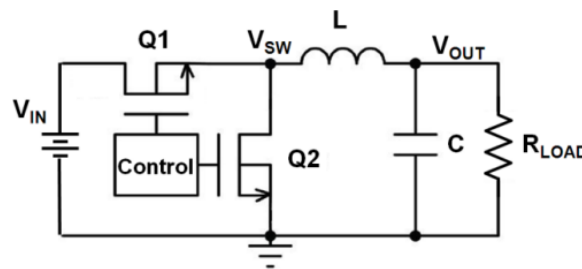


Figure 1.1: synchronous buck converter

The circuit mainly consists of two MOSFETs, an inductor, a capacitor, and a PWM controller. The high-side MOSFET connects the input voltage to the inductor when it is turned on, while the low-side MOSFET provides a return path for the current when the high-side device is turned off. Both switches are driven by complementary PWM signals generated by a driver circuit such as the IR2101. When the high-side MOSFET is turned on, the input voltage is applied across the inductor. The inductor current rises linearly, and energy is stored in its magnetic field while supplying current to the load. When the high-side switch is turned off, the low-side MOSFET turns on immediately to maintain the inductor current. During this interval, the inductor releases

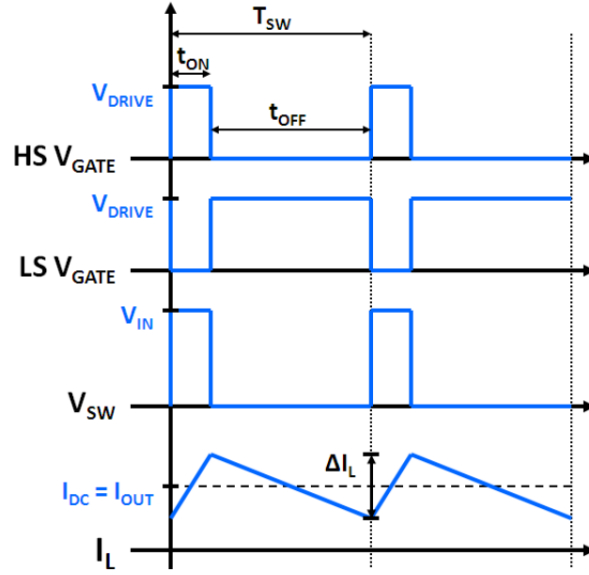


Figure 1.2: Waveform of synchronous buck converter

the stored energy to the load, and the current through it decreases gradually. This continuous transfer of energy between the inductor and the load results in a smooth DC output voltage after filtering by the capacitor.

$$D = \frac{V_{OUT}}{V_{IN}} \quad (1.1)$$

$$L_{MIN} = \frac{(V_{IN} - V_{OUT}) \cdot D}{LIR \cdot I_{OUT,MAX} \cdot f_{SW}} \quad (1.2)$$

$$\Delta V_{PP} = CVR \cdot V_{OUT} \quad (1.3)$$

$$ESR_{MAX} = \frac{\Delta V_{PP}}{I_{OUT,MAX}} = \frac{CVR \cdot V_{OUT}}{I_{OUT,MAX}} \quad (1.4)$$

$$C_{MIN} = \frac{LIR \cdot I_{OUT,MAX}}{8 \cdot f_{SW} \cdot CVR \cdot V_{OUT}} \quad (1.5)$$

$$I_{PEAK} = I_{OUT,MAX} + \frac{\Delta I_L}{2} \quad (1.6)$$

These equations are used for designing the main components of a synchronous buck converter. The duty cycle determines the voltage conversion ratio, while the inductance and capacitance values are chosen based on ripple current and voltage requirements. The ESR and peak current equations

help ensure stability, efficiency, and safe operation of the converter under load conditions

The output voltage of the converter depends on the duty cycle of the PWM signal, which is the ratio of the ON-time of the high-side MOSFET to the total switching period.

why synchronous buck converter:

The synchronous buck converter provides higher efficiency and lower heat dissipation compared to a non-synchronous version, since both switches are actively controlled and the conduction losses are minimized. It is commonly used in applications such as microcontroller power supplies, motor drivers, and battery-powered systems where efficiency and reliability are critical

1.3 C2000 Microcontroller Overview

Texas Instruments C2000 microcontrollers are specifically optimized for low-latency, precision control tasks that make them highly suitable for power electronics, motor drives, solar inverters, and automotive power stages. Advanced analog and digital peripherals are tightly integrated with high-speed CPUs operating up to 200 MHz for efficient execution of complex algorithms. Floating and fixed-point math units allow for advanced signal processing and control calculations.

Hardware accelerators and on-chip peripherals support fast ADC-CPU-PWM interactions, real-time protection, and multi-phase/multilevel power topologies. Safety features include lock-step CPU operation, error checking (ECC protection), fault detection, and secure hardware interfaces.

C2000 MCUs provide pin-to-pin and code compatibility across the device family to support scalable product designs. Typical applications are buck/boost converters, shunt current sensing, motor control (TT/PMSM/ACIM), and renewable energy systems.

1.4 Code Composer Studio (CCS) Overview

Code Composer Studio is the official TI integrated development environment for Texas Instruments embedded processors, including all C2000 microcontrollers. CCS provides a suite of tools for code development, debugging, profiling, and system analysis.

Built on top of the Eclipse open-source framework since version 4.0 and upgraded to Eclipse Theia since v20, CCS offers extensibility and intuitive interfaces.

It includes C/C++ compiler optimization, graphical debugging and real-time monitoring, and code analysis tools.

CCS supports JTAG programming, hardware-based debugging, and simulation for embedded system workflows.

Developers can create projects step by step, starting from conceptual design to source code development, configuration, debugging, and up to system analysis.

Cloud-based versions of CCS are available to support remote development and resource management.

1.5 EPWM Module Overview

The ePWM module in the C2000 microcontroller is a high-performance peripheral that generates precise PWM signals, controlling electronic power devices such as MOSFETs or IGBTs. It allows switching frequency, duty cycle, and phase to be set with high accuracy — ideal for applications such as DC-DC converters and motor control.

The ePWM module includes eight submodules that allow for extensive PWM functionality:

- **Time-base:** Provides period and frequency control using a 16-bit counter.
- **Counter Compare:** Sets the duty cycle based on a comparison of the counter values with preset thresholds.
- **Action Qualifier:** Determines the output behavior on occurrences of counter matches: set, clear, toggle.
- **Dead-band:** Inserts rising and falling edge delays to protect power devices in complementary outputs.
- **PWM Chopper:** Modulates PWM signals for power reduction or EMI control.
- **Event Trigger:** Provides triggers to ADC or other peripherals that are synchronized with PWM.
- **Trip-zone:** Detects fault conditions and safely disables PWM outputs.
- **Digital Compare:** Compares digital inputs for triggering PWM events.

Using ePWM in CCS for C2000 MCUs

In CCS, there are two ways to configure the ePWM module: direct programming of the registers and SysConfig, an integrated graphical configuration tool for easy setup. Key configurations include:

- Setting PWM frequency via the time-base period register.
- Defining the duty cycle through counter compare registers.

- Configuration of dead-band timings for safe complementary output switching.
- Synchronizing several PWM modules for multiphase applications, using sync-in/out signals.
- Implementing trip-zone protection modes for fault tolerance.

1.6 Objective of the Project

The main goal of this project is to design and build a synchronous buck converter that steps down a 12 V DC supply to a steady 5 V output for efficiently driving a 5 V TT motor. The converter is controlled using PWM signals from a C2000 microcontroller, which helps maintain accurate voltage regulation and high efficiency. This design ensures smooth motor operation, lower power loss, and better overall control performance.

CHAPTER 2

Modeling and stimulation of Buck-converter

2.1 Introduction

In this simulation, synchronous buck converter has been modeled using Simulink to regulate a DC voltage source. The setup includes high-side and low-side MOSFETs driven by PWM signals to perform the step-down conversion. An inductor and capacitor form the output filter to smooth the voltage and current delivered to the load. The simulation also includes a voltage measurement block to monitor the output, demonstrating how the converter reduces a 12V input to a lower DC output (around 6V in this case). This model allows the study of converter behavior under varying load conditions and provides insight into voltage regulation, ripple, and system dynamics.

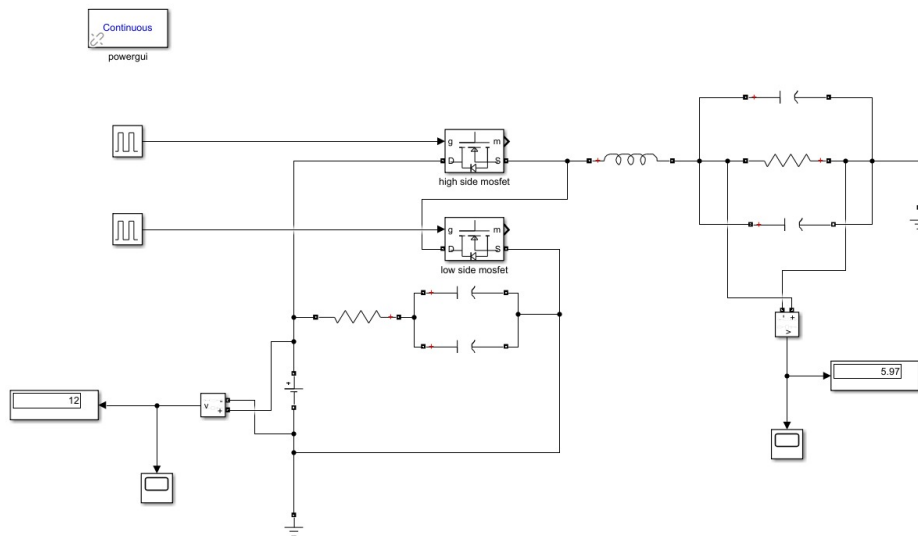


Figure 2.1: MATLAB simulation.

2.2 input and output

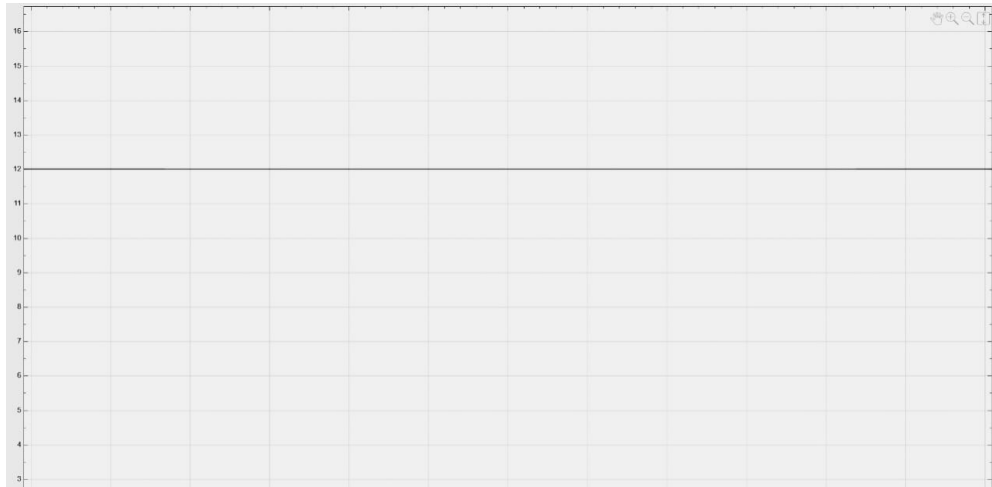


Figure 2.2: Input voltage to buck converter.



Figure 2.3: Output voltage during buck converter startup (MATLAB simulation).

The first plot displays a flat, steady input voltage supplied to the converter, confirming a stable test condition without noise or disturbances.

The second plot shows the output voltage of the buck converter during startup, with high-frequency oscillations that quickly damp out, indicating fast settling and good system stability.

2.3 System Modeling

This part explains how the circuit steps down a higher DC voltage to a lower DC voltage. In this open-loop setup, the C2000 microcontroller generates a fixed-duty PWM signal that drives the two MOSFETs in a complementary way. The converter does not use any feedback control, so the output voltage depends on the input voltage and the duty ratio applied from the controller.

Circuit Description

Input voltage = 12V

output voltage = 5V

MOSFET = High-side , Low-side

Inductor = 33 μ H

Output capacitor = 47 μ F

Load – represents the connected device(motor)

ASSUMPTIONS

$f_s = 100$ kHz -Switching frequency

$I_{out} = 3$ A -Load current

$\Delta I_L = 0.9$ A -Inductor current ripple (30% of output current)

$\Delta V_{C,ideal} = 25$ mV -Allowable output voltage ripple

FORMULA

1. Duty Cycle: $D = \frac{V_{out}}{V_{in}} = \frac{5}{12} = 0.4167$

2. Inductor Value: $L = \frac{(V_{in} - V_{out}) D}{\Delta I_L f_s} = \frac{(12 - 5) \times 0.4167}{0.9 \times 100 \times 10^3} = 32.4 \mu\text{H}$

3. Capacitor Value: $C = \frac{\Delta I_L}{8 f_s \Delta V_{C,ideal}} = \frac{0.884}{8 \times 100 \times 10^3 \times 0.025} = 44.2 \mu\text{F}$

The capacitor value is rounded to the standard value 47 μ F.

The inductor value is rounded to the standard value 33 μ H.

2.4 System Modeling Illustrations

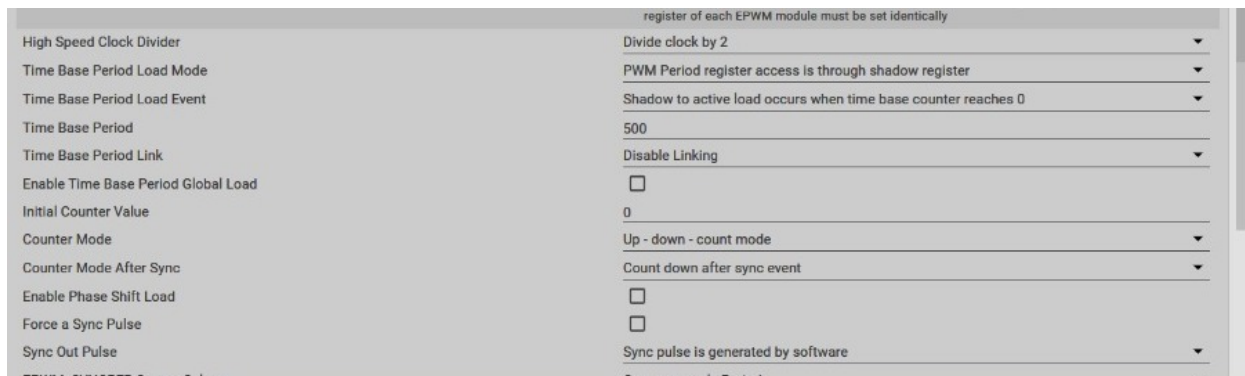


Figure 2.4: System configuration-1

This image shows the time base and period configuration for the ePWM module. Here, the high-speed clock divider is selected as divide by 2, with a time-base period of 500. The direction of the counter is selected as up-down, such that PWM pulses are generated symmetrically, leading to a better quality of output waveform with low harmonic distortion. These settings are quite vital for the specification of PWM frequency and enabling converter operation in stable modes. This

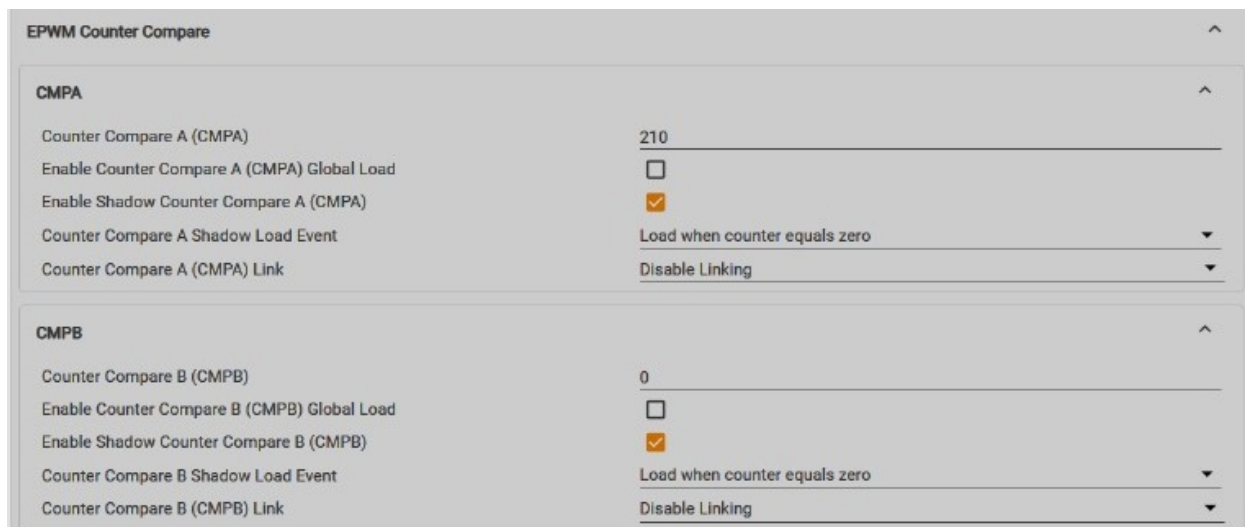


Figure 2.5: System configuration-2

figure shows the ePWM counter compare settings: $CMPA = 210$ and $CMPB = 0$; the shadow register for CMPA is enabled, loading the value at counter zero. This will define the PWM pulse width and enable fine control of the output voltage or current for converter applications, making this register setting very important for setting precise duty cycles.

ePWMxA Output Configuration

ePWMxA Global Load Enable

ePWMxA Shadow Mode Enable

ePWMxA One-Time SW Force Action

No change in the output pins

ePWMxA Continuous SW Force Action

Software forcing disabled

ePWMxA Event Output Configuration

ePWMxA Time base counter equals zero

No change in the output pins

ePWMxA Time base counter equals period

No change in the output pins

ePWMxA Time base counter up equals COMPA

Set output pins to High

ePWMxA Time base counter down equals COMPA

Set output pins to low

ePWMxA Time base counter up equals COMPB

No change in the output pins

ePWMxA Time base counter down equals COMPB

No change in the output pins

ePWMxA T1 event on count up

No change in the output pins

ePWMxA T1 event on count down

No change in the output pins

ePWMxA T2 event on count up

No change in the output pins

ePWMxA T2 event on count down

No change in the output pins

Figure 2.6: System configuration-3

This image documents the output event configuration for the ePWM module. It shows the change in the behavior of the output pin when a time base counter matches COMPA both at high and low while ensuring accurate pulse generation. This level of configuration is critical in ensuring synchronization of external switching devices.

2.5 C2000 PWM Driverlib Code

Listing 2.1: PWM Code for Synchronous Buck Converter

```
1 //
2 // Included Files
3 //
4 #include "driverlib.h"
5 #include "device.h"
6 #include "board.h"
7 #include "c2000ware_libraries.h"
8 // Desired duty cycle (42)
9 #define DUTY_CYCLE 0.42f
10
11 void initEPWM1Duty(void)
12 {
13     // Get the time-base period of ePWM1
14     uint32_t tbprd = EPWM_getTimeBasePeriod(EPWM1_BASE);
15
16     // Set PWM duty cycle (EPWM1A)
17     EPWM_setCounterCompareValue(EPWM1_BASE, EPWM_COUNTER_COMPARE_A,
18         DUTY_CYCLE * tbprd);
19 }
20
21 void main(void)
22 {
23     Device_init();
24     Device_initGPIO();
25     Interrupt_initModule();
26     initEPWM1Duty();
27     Interrupt_initVectorTable();
28     Board_init();
29     C2000Ware_libraries_init();
30
31     EINT;
32     ERTM;
33
34     while(1) {}
35 }
36 //
37 // End of File
38 //
```

CHAPTER 3

Hardware implementation of project

3.1 Introduction

This chapter explains the design and implementation of a Buck Converter using KiCad. The converter is developed to step down the DC voltage efficiently for use in solar-powered applications. The schematic and PCB layout are designed in KiCad, and the layout is converted into G-code for fabrication.

3.2 Steps involved designing a PCB

3.2.1 Drawing the schematic diagram

The complete circuit diagram is drawn using KiCad Schematic Editor, Eeschema. All required components such as resistors, capacitors, ICs, and connectors are selected from the KiCad library or custom libraries.

3.2.2 Assigning footprints to components

Each component is allocated an appropriate footprint, representing the actual size and configuration of the physical pin arrangement of the part, when the schematic has been completed.

3.2.3 ERC

An ERC is run to check the schematic for such common errors including, but not limited to unconnected pins, shorted nets, and incorrect power connections.

3.2.4 Generating the netlist

Once the schematic has been checked, a netlist is created. It is a file that includes data about all the electrical connections, used to associate the schematic with the layout of the PCB.

3.2.5 Designing the PCB layout

The board outline in KiCad PCB Editor, also known as PCBNew, is specified, and all components are placed with regard to design requirements, including the flow of signals, paths of power, and thermal management.

3.2.6 Routing the traces

The copper traces are routed between the component pads to complete the electrical connections; proper trace width and clearance are maintained to handle the required current and avoid interference. The track width used is 1mm and the minimum clearance is 0.5mm.

3.2.7 DRC

A Design Rule Check is done to ensure all design constraints that include minimum trace width, spacing, hole sizes are met and there are no errors left within the layout.

3.2.8 Generating gerber and drill files

The final step is to generate the Gerber files, drill files, and BOM. These files are needed by the PCB manufacturer for fabrication and assembly.

3.3 Schematic

The schematic shows the design of a synchronous buck converter used to step down a higher DC voltage to a lower level efficiently. In this circuit, the IR2104 gate driver controls two MOSFETs—IRFZ44N as the high-side switch and IRF540N as the low-side switch. The gate driver receives PWM signals from the controller and drives the MOSFETs in a proper switching sequence to regulate the output voltage. The inductor (L1) and capacitors (C3 and C4) work together to smooth the output and reduce voltage ripple. Additional capacitors (C1, C2, and C5) help stabilize the driver's supply voltage, while resistors (R1, R2, R3, and R10) limit the gate current and provide proper biasing. The final output, labeled BUCK_Vout, provides a clean and stable DC voltage that can be used to power a motor or other DC loads.

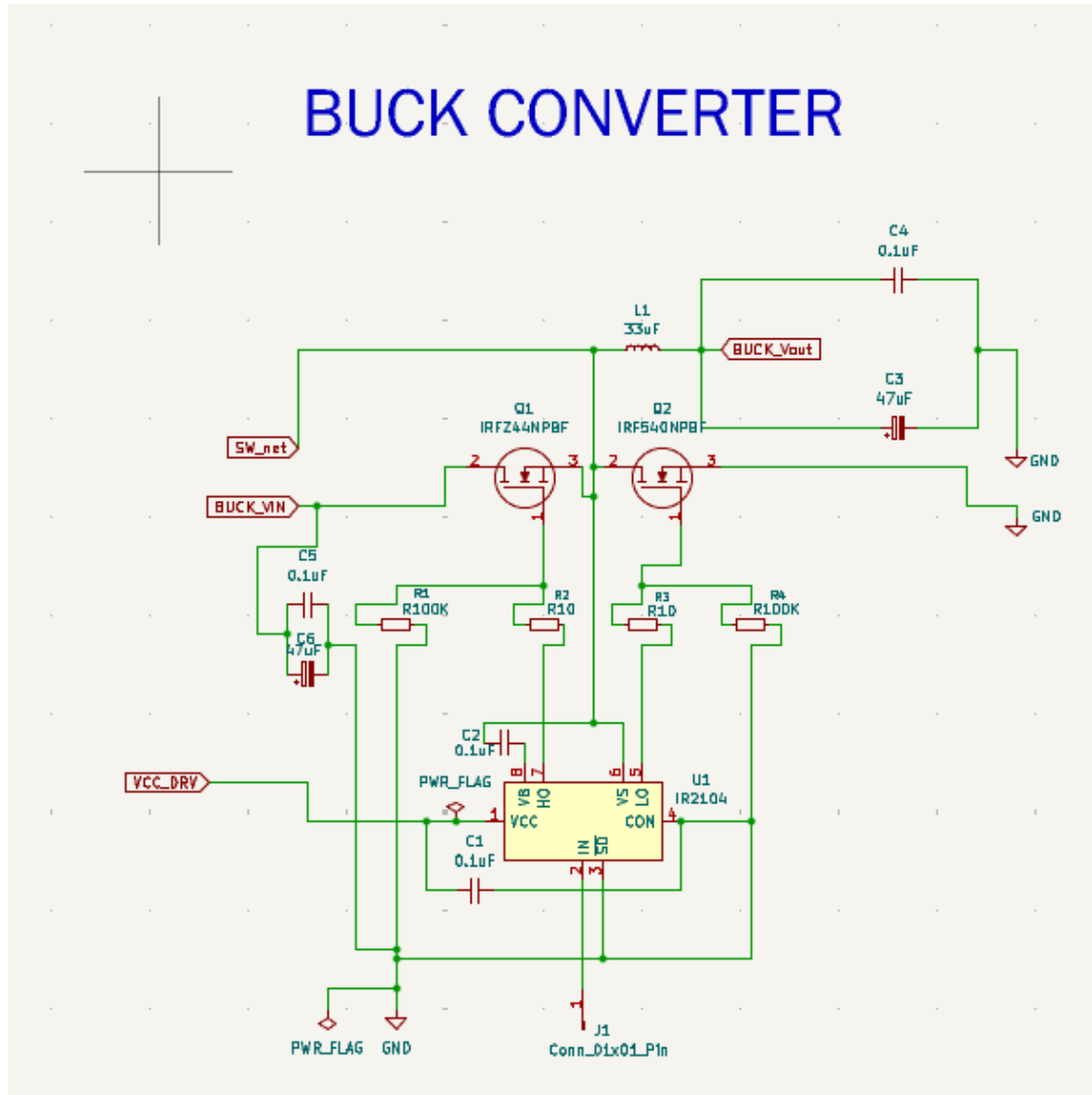


Figure 3.1: schematic of buck converter.

3.4 footprints and BOM

The components used in this project were carefully selected based on performance requirements and availability. All parts, including semiconductors, passive components, and connectors, were procured from Element14 India. This ensures that the materials are genuine and meet industrial standards, enhancing the reliability and reproducibility of the design.

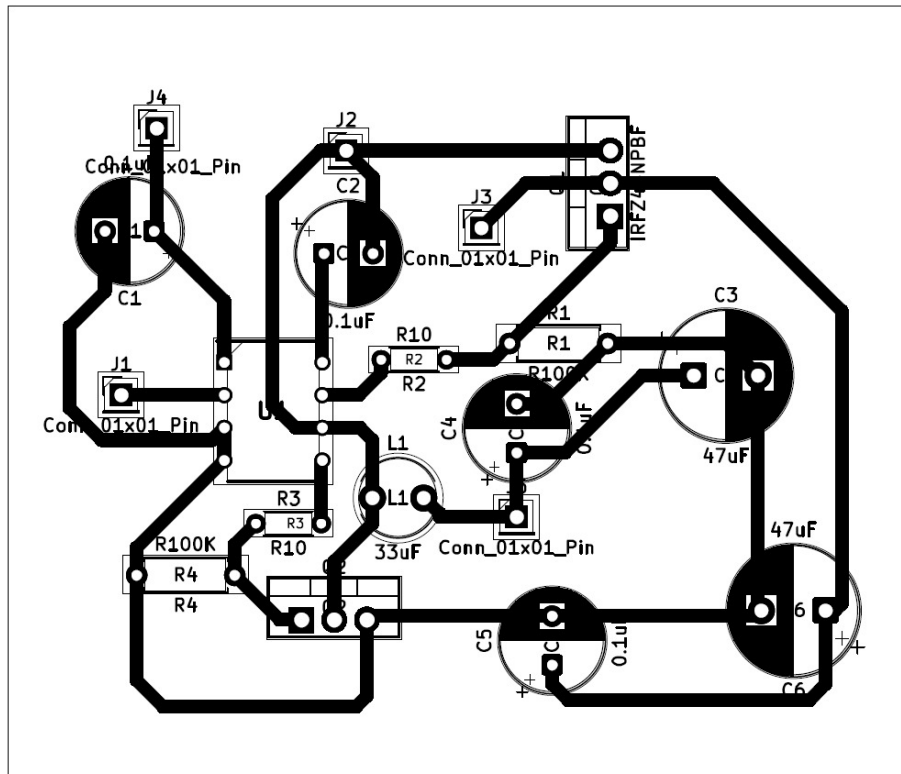
Symbol : Footprint Assignments			
1	C1 -	0.1uF :	Capacitor_THT:CP_Radial_D4.0mm_P1.50mm
2	C2 -	0.1uF :	Capacitor_THT:CP_Radial_D4.0mm_P1.50mm
3	C3 -	47uF :	Capacitor_THT:CP_Radial_D10.0mm_P5.00mm
4	C4 -	0.1uF :	Capacitor_THT:CP_Radial_D4.0mm_P1.50mm
5	C5 -	0.1uF :	Capacitor_THT:CP_Radial_D4.0mm_P1.50mm
6	C6 -	47uF :	Capacitor_THT:CP_Radial_D10.0mm_P5.00mm
7	J1 -	Conn_01x01_Pin :	Connector_PinHeader_1.00mm:PinHeader_1x01_P1.00mm_Vertical
8	L1 -	33uF :	Inductor_THT:L_Radial_D6.0mm_P4.00mm
9	Q1 -	IRFZ44NPBF :	Package_TO_SOT_THT:TO-220-3_Vertical
10	Q2 -	IRF540NPBF :	Package_TO_SOT_THT:TO-220-3_Vertical
11	R1 -	R100K :	Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal
12	R2 -	R10 :	Resistor_THT:R_Axial_DIN0204_L3.6mm_D1.6mm_P5.08mm_Horizontal
13	R3 -	R10 :	Resistor_THT:R_Axial_DIN0204_L3.6mm_D1.6mm_P5.08mm_Horizontal
14	R4 -	R100K :	Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm_Horizontal
15	U1 -	IR2104 :	foot_prints:DIP762W56P254L988H533Q8N

Figure 3.2: foot print.

S. N	Name of the component	MPN	Order code	MOQ	Required quantity	Unit cost	Cost
1	Capacitors	MCMHR50V104M4X7	1871011	1	5	8.45	42.25
2	Capacitors	MCKSK100M470G17S	2610810	1	2	22.51	45.02
3	Pin header	MC34637	1675769	1	2	10.29	20.58
4	Inductor	RLB0914-330KL	2309242	1	2	31.32	62.64
5	MOSFET(Q1)	IRFZ44NPBF	8650225	1	2	125.25	250.5
6	Resistors	MFR-25FTE52-100K	4064087	1	2	0.835	1.67
7	Resistors	CFR25SJT-52-10R	3496828	10	10	1.75	17.5
8	Gate Driver	IR2104PBF	8638748	1	2	308.33	616.66
9	MOSFET(Q2)	IRF540NPBF	8648298	1	2	90.67	181.34

Figure 3.3: bill of material.

3.5 Layout



3.6 Fabricated PCB

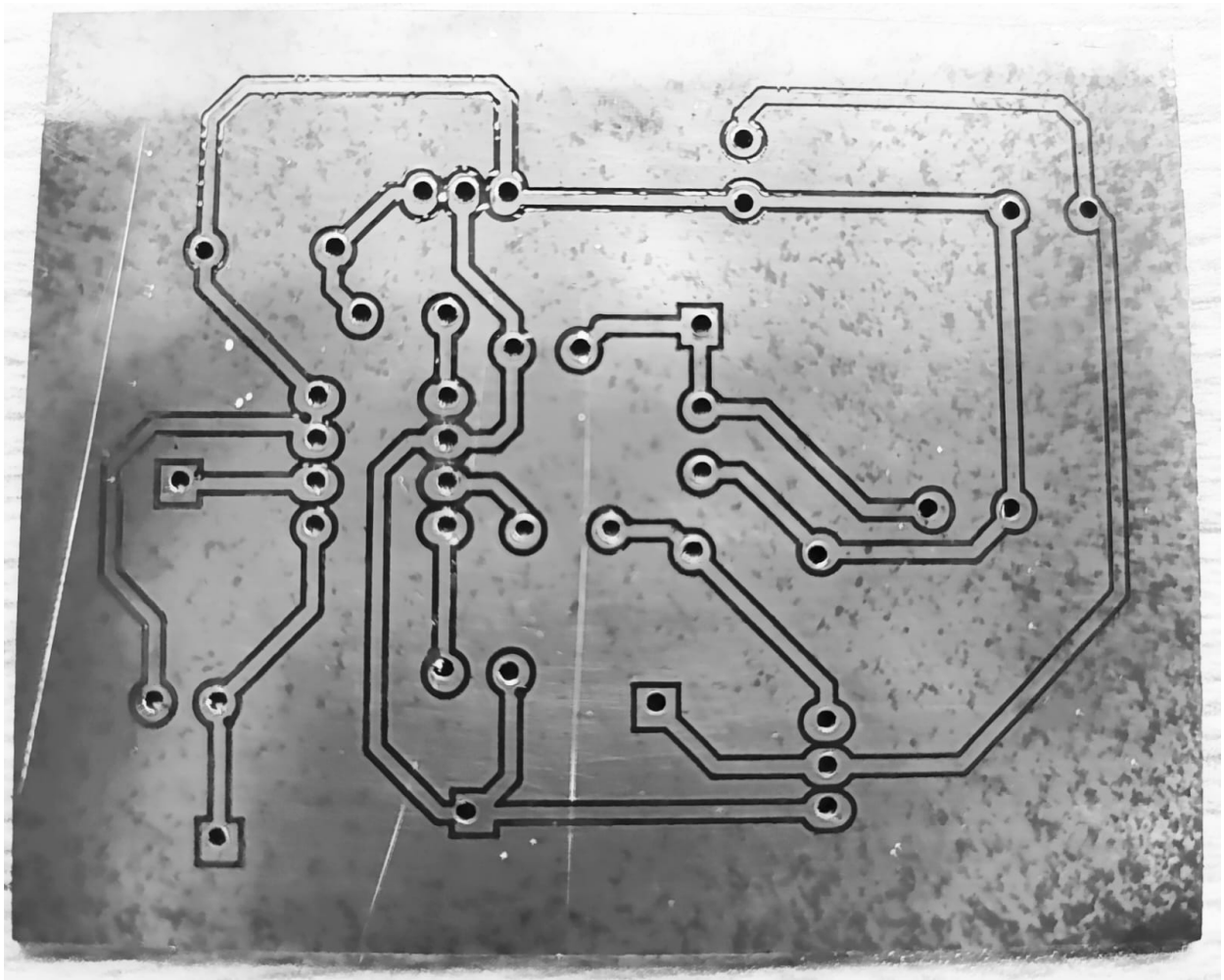


Figure 3.4: Fabricated PCB layout of buck converter

After the PCB layout was designed in KiCad, the design files were given to the fabrication section. The copper pattern was first transferred onto the copper-clad board. Then, the board was passed through the etching process to remove unwanted copper and form the tracks according to the layout. The drilling machine was used to make holes for mounting the components. After drilling, edge cutting was done to shape the PCB to its required dimensions. Finally, the board was cleaned and tested for continuity and short circuits to ensure that all tracks were correctly formed and isolated.

3.7 Final out of PCB

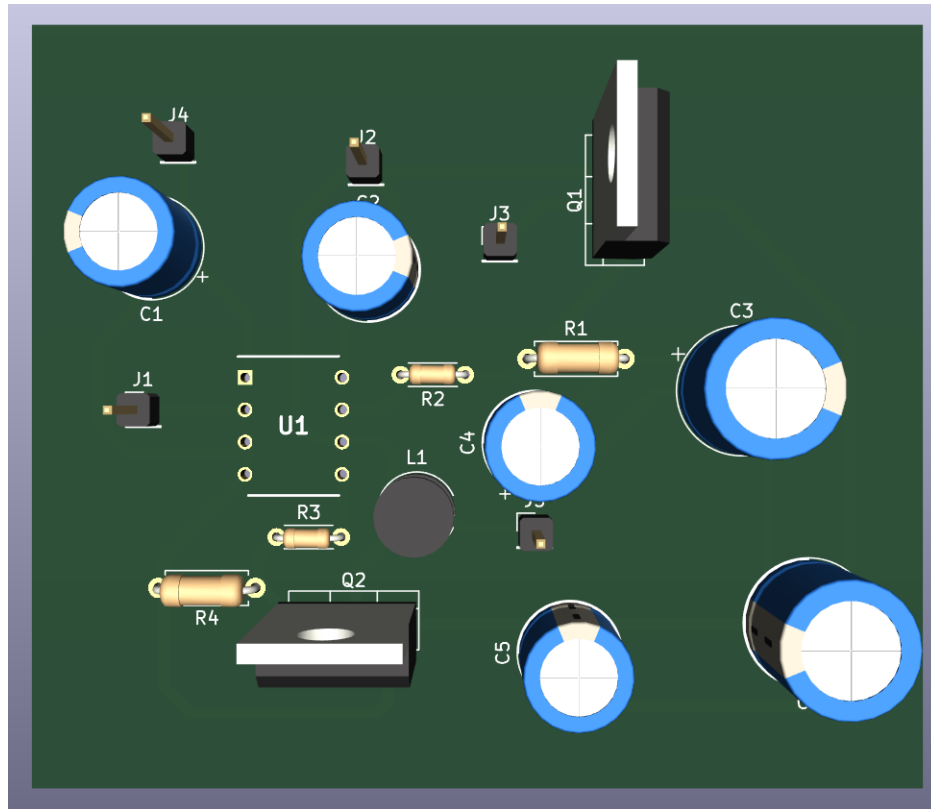


Figure 3.5: 3Dview of buck covnerter

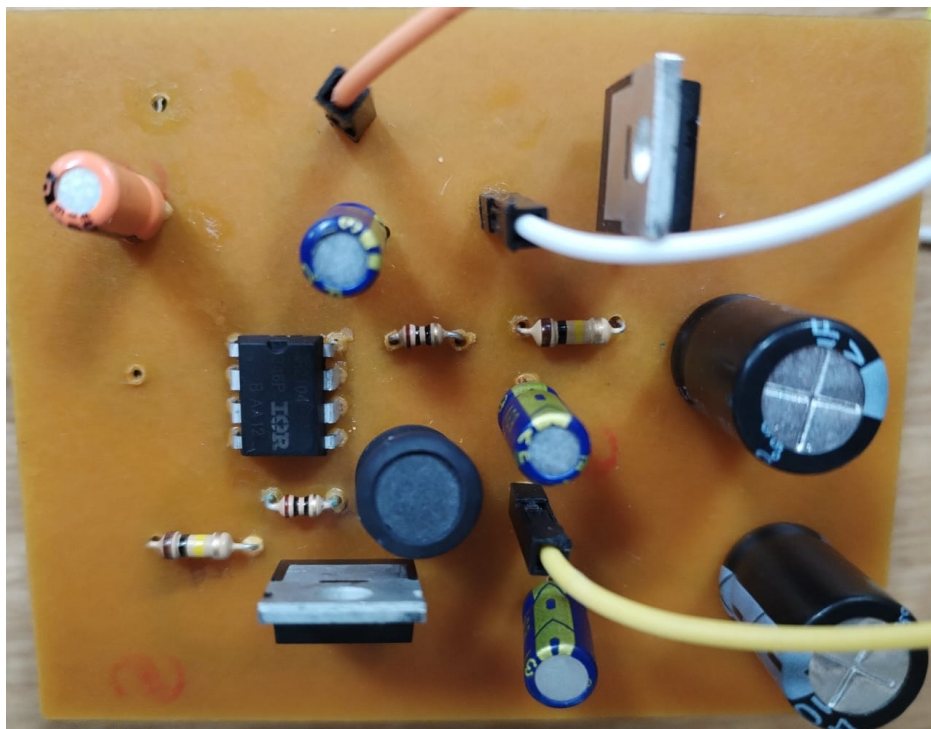


Figure 3.6: Assembled PCB Board of buckconverter

CHAPTER 4

Conclusion

This project successfully designed and implemented an open-loop synchronous buck converter controlled by a TI C2000 microcontroller to step down a 12 V DC supply to 5 V for driving a TT motor. The open-loop configuration demonstrated efficient voltage regulation through PWM duty cycle control, verified by simulation and hardware testing. The integration of the IR2104 gate driver and complementary MOSFET switching ensured stable and reliable operation with low power losses.

This work provided practical experience in power electronics design, PCB fabrication, and embedded PWM generation using Code Composer Studio and DriverLib for open-loop control.

SIMILARITY CHECK REPORT

Report: Team5

Team5

by D Uma

General metrics

4,966	788	90	3 min 9 sec	6 min 3 sec
characters	words	sentences	reading time	speaking time

Score

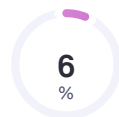


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Writing Issues

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